

Unit 9 – Inference for Quantitative Data: Slopes

Notes 2 – Significance Testing for Slope

The significance test for a slope is called the _____. It can help us answer three different questions with the hypotheses:

Is the relationship between the explanatory and response variable negative? $H_0: \beta = 0$ $H_a: \beta < 0$	Is there a relationship between the explanatory and response variable? $H_0: \beta = 0$ $H_a: \beta \neq 0$	Is the relationship between the explanatory and response variable positive? $H_0: \beta = 0$ $H_a: \beta > 0$
---	---	---

The conditions for a significance test are the same conditions for a confidence interval:

Linear	Independent	Normal	Equal SD	Random
A scatterplot of the data shows a linear pattern	When sampling, check the 10% condition	A graph of the residuals shows an approximately normal distribution	The residual plot shows random scatter	The data is collected via a random sample or obtained via random assignment

To create your test statistic and find your p-value:

--	--

In your conclusion, you would state the results of your significance test (reject or fail to reject) and the interpret the findings in context

Computer Output

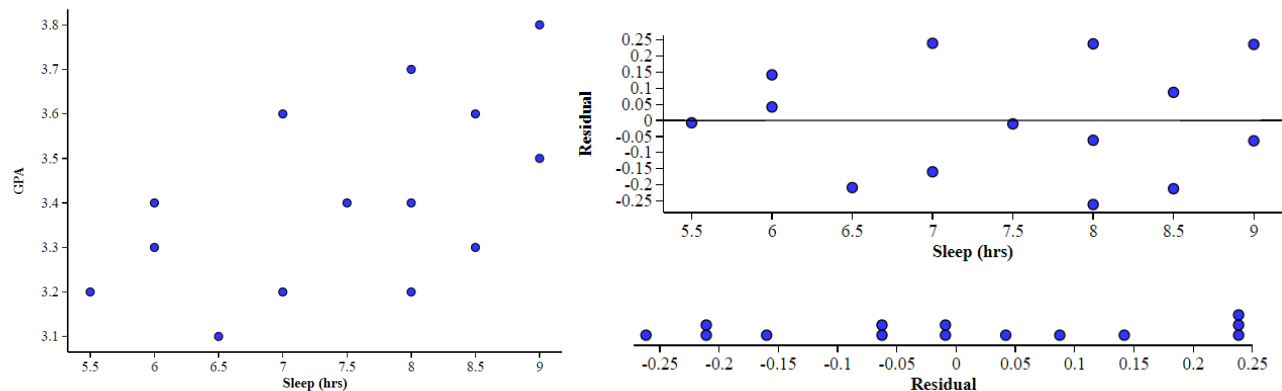
Predictor	Coef	St Dev	t ratio	P
Constant	-12.56	5.222	2.405	.0370
IQ	0.9043	0.6678	1.354	.2055

s = 6.143 R-sq = 47.7% R-sq(adj) = 44.8%

Ex) A school counselor is worried about how much sleep their students are getting every night. He is concerned that the number of hours of sleep that his students get each night is affecting their GPA in a negative way. He selects a random sample of 14 seniors in his district and asks them how many hours of sleep they get on a typical school night. He then uses school records to determine the most recent grade-point average (GPA) for each student. His data are given below.

Sleep (hours)	9	8.5	9	7	7.5	6	7	8	5.5	6	8.5	6.5	8	8
GPA	3.8	3.3	3.5	3.6	3.4	3.3	3.2	3.2	3.2	3.4	3.6	3.1	3.4	3.7

Scatterplot



Equation	n	r^2	s	S_x
$\hat{y} = 2.65 + 0.102x$	14	0.313	0.18	1.15

Do these data provide convincing evidence of a linear relationship between the hours of sleep students typically get and their academic performance, as measured by their GPA? Carry out a significance test at the $\alpha = 0.05$ level.

Calculator - T Test for Slope

F: LinRegTTest

XList:

YList:

Freq: 1

β & ρ : $\neq 0$ < 0 > 0

RegEq:

Ex) The computer output given at the right shows a regression analysis of an honors social science course (score in points) versus a reading comprehension score (in points) for 25 sophomores at your school.

$$\text{Social science score} = 76.56 + 0.731\text{Reading}$$

Predictor	Coef	SE Coef	T	P
Constant	76.56	10.168	7.53	<.0001
Reading	0.731	0.0351	20.84	<.0001

$$s = 25.83 \quad R\text{-Sq} = 0.610 \quad R\text{-Sq(Adj)} = 0.610$$

- a) Identify the value of the standard deviation of the residuals and interpret this value in the context of the problem.
- b) Identify the value of the estimated standard deviation of the slope and interpret this value in the context of the problem.
- c) Carry out a hypothesis test for these data and interpret your results in the context of the problem. You may assume that the necessary assumptions are satisfied.